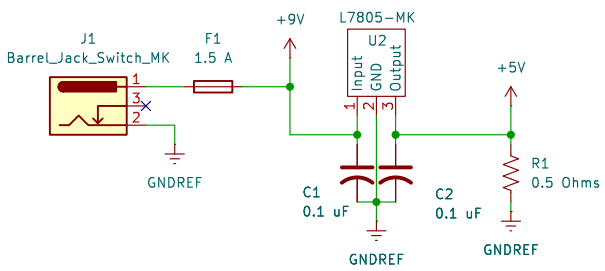
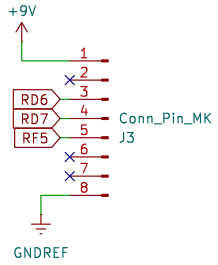
Power supply is not needed, however is here for testing purposes, but the main power will be jumped from Kelton (main board).

Power Supply Circuit



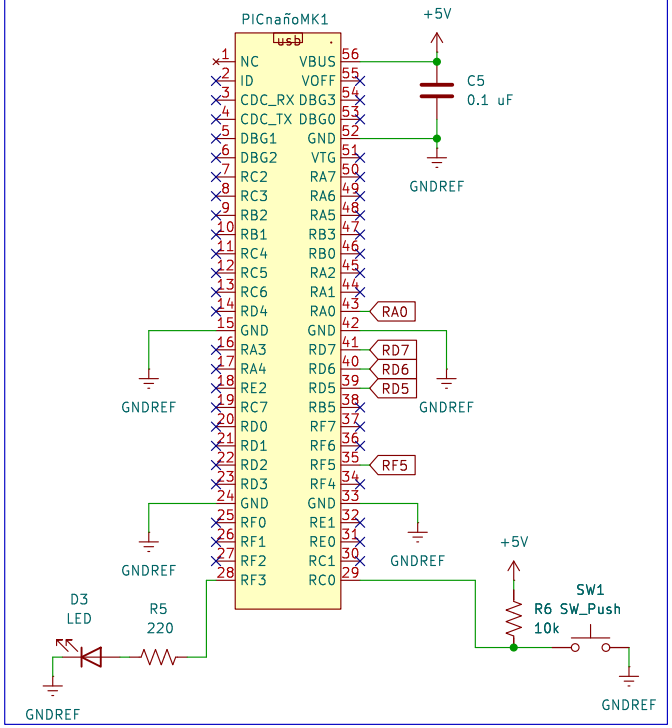
In connector circuit, two PWM pins RD6 and RD7 connect to Microcontroller and RF5 is debugging connection. The group will run a shared power supply of 9v, also we have common ground.

Connector Circuit



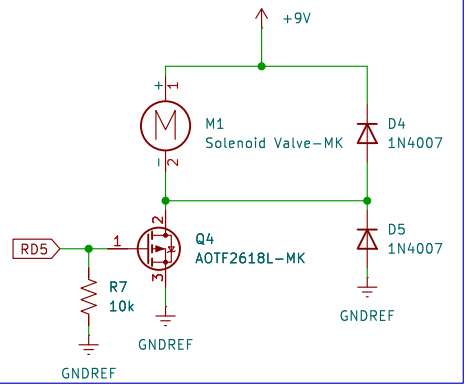
Microchip microcontroller recieves signals from keltons board and reroutes them to the solenoid and speaker. RF5 is a debug as well as the led and push button.
(may need to add more debug points or empty spaces in case of failed connections.)

MicroController Circuit



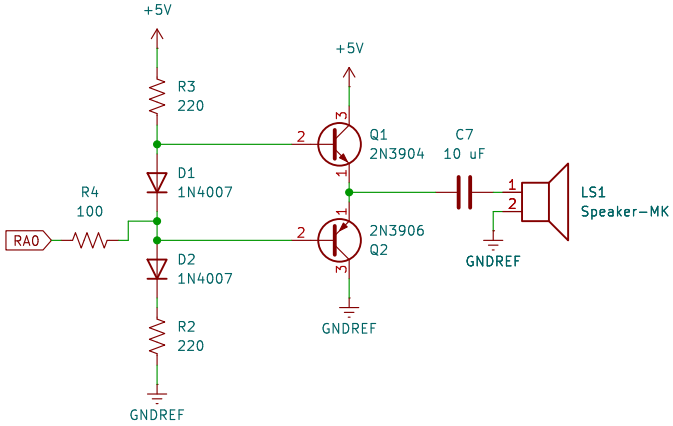
The solenoid recieves the signal from RD5 and will converted from microchips RD6 pin. So RD6 is recieved from Keltons board and converted to RD5 to open the solenoid. The diodes are a freewheeling diode. A MOSFET is also encorporated for the solenoid.

Solenoid Valve Circuit



The speaker recieves an analog input from the microchip to play multiple sounds. The microchip takes a signal from RD7 and reroutes it into analog from RA0 and pushes through the speaker circuit to play sound.

Speaker Circuit



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